

Numerical example.

	x_1	x_2	y
(a)	1	2	3
(b)	2	1	4
(c)	3	3	5

$n=3$ Find $\theta_0, \theta_1, \theta_2 = 0$.
 $d=2$ $0 \ 0 \ 0$ — initially to 0.
 $x = \mathbb{R}^{d+1}$

$$h = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

assume $x_0 = 0 + 1$

$$\text{Find } \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

Apply gradient descent and compute new value of θ (vector) after two iterations.

init θ_0 with 0.

$$x_1 + x_2 \rightarrow \hat{y}$$

x_0	x_1	x_2	y
1	1	2	3
1	2	1	4
1	3	3	5

$m=3$ (no. of training examples)

$x_0 \rightarrow$ to include intercept term

$\theta_0 \rightarrow$ intercept term

$\theta_1 \rightarrow$ parameter of house area x_1

$\theta_2 \rightarrow$ parameter of x_2

} unknown

hypothesis function (prediction)

3 values/sample.

$$h_\theta(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

Since $x_0 \rightarrow 1$

$$h_\theta(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2$$

For each house sample prediction

$$(a) \theta_0 + \theta_1(1) + \theta_2(2)$$

$$(b) \theta_0 + \theta_1(2) + \theta_2(1)$$

$$(c) \theta_0 + \theta_1(3) + \theta_2(3)$$

Cost function (how wrong prediction)

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_\theta(x^{(i)}) - y^{(i)})^2$$

$$J(\theta) = \frac{1}{2} \left[(\theta_0 + \theta_1 + 2\theta_2 - 3)^2 + (\theta_0 + 2\theta_1 + \theta_2 - 4)^2 + (\theta_0 + 3\theta_1 + 3\theta_2 - 5)^2 \right]$$

Find $\theta_0, \theta_1, \theta_2$ that minimizes $J(\theta)$

Gradient Descent Formula

for each parameter.

$$\theta_j := \theta_j - \alpha \frac{\partial J}{\partial \theta_j}$$

α — learning rate.

step1: compute partial derivative for

θ_0, θ_1 and θ_2

for θ_0

$$\frac{\partial J}{\partial \theta_0} = \sum_{i=1}^4 (h_0(x^{(i)}) - y^{(i)})$$

→ initialise parameter.

$$\theta_0 = 0, \theta_1 = 0, \theta_2 = 0$$

for $\theta_1 =$

$$\frac{\partial J}{\partial \theta_1} = \sum_{i=1}^4 (h_0(x^{(i)}) - y^{(i)}) x_1^{(i)}$$

for $\theta_2 =$

$$\frac{\partial J}{\partial \theta_2} = \sum_{i=1}^4 (h_0(x^{(i)}) - y^{(i)}) x_2^{(i)}$$

first iteration

when θ_0, θ_1 and $\theta_2 = 0$

$$h_0(x) = 0$$

$$h_0(x) = 0(x_0) + 0(x_1) + 0(x_2)$$

$$h_0(x) = 0$$

$$\text{error} = h - y$$

$$(a) 0 - 3 = -3$$

$$(b) 0 - 4 = -4$$

$$(c) 0 - 3 = -3$$

Gradient Formula.

(1) 2

Iteration 1 $\alpha = 0.1$

$$\theta_0 = 0, \theta_1 = 0, \theta_2 = 0 \quad \text{updated } \theta = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \theta_2 \end{bmatrix}$$

hypothesis function

$n=3$

$$h_0(x) = \theta_0 x_0 + \theta_1 x_1 + \theta_2 x_2$$

$$\text{beg } x_0 \Rightarrow 1$$

$$h_0(x) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 \quad \text{--- (1)}$$

for each sample prediction

$$h_0(x) = \theta_0 + \theta_1(1) + \theta_2(2)$$

$$h_0(x) = \theta_0 + 2\theta_1 + 1\theta_2$$

$$h_0(x) = \theta_0 + 3\theta_1 + 3\theta_2$$

sample
 x_2
parameters

objective function.

$$i \rightarrow 1 \text{ to } n \rightarrow \text{no. of samples}$$

$$j = 1 \text{ to } d \rightarrow \text{no. of features}$$

$$J(\theta) = \frac{1}{2} \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)})^2$$

$$\text{for } \theta_0 \quad \theta_0 := \theta_0 - \alpha \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)})$$

$$\theta_j := \theta_j - \alpha \sum_{i=1}^n (h_0(x^{(i)}) - y^{(i)}) x_j^{(i)}$$

$$\sum_{i=1}^3 h_0(x^{(i)}) = (\theta_0 + \theta_1 + \theta_2 - 3) \times 1$$

$$i=2 \quad h_0(x^{(2)}) = (\theta_0 + 2\theta_1 + \theta_2 - 4) \times 1$$

$$i=3 \quad h_0(x^{(3)}) = (\theta_0 + 3\theta_1 + 3\theta_2 - 5) \times 1$$

$$0 - 3 \times 1 = -3$$

$$-4 \times 1 = -4$$

$$-5 \times 1 = -5$$

$$-12$$

$$\theta_0 = 0.1(-12)$$

$$0 - (1.2) = \underline{\underline{-1.2}}$$

$$\theta_0 = 1.2$$

$$\theta_1 = 2.6$$

$$\theta_2 = 2.5$$

$$\theta_0 = -1.02$$

$$\theta_1 = -2.41$$

$$\theta_2 = -2.6$$

Iteration 2

$$\theta_0 = 1.2, \theta_1 = 2.6, \theta_2 = 2.5$$

$$\theta_0 = \theta - \alpha \sum_{i=1}^3 (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_1^{(i)}$$

$$\theta_1 = \theta - \alpha \sum_{i=1}^3 (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_1^{(i)}$$

$$\sum_{i=1}^3 (h_{\theta}(x^{(i)}) = (\theta_0 + 1\theta_1 + 2\theta_2) - 3 = -3 \times 1 = -3$$

$$h_{\theta}(x^{(2)}) = (\theta_0 + 2\theta_1 + 1\theta_2) - 4 = -4 \times 2 = -8$$

$$h_{\theta}(x^{(3)}) = (\theta_0 + 3\theta_1 + 3\theta_2) - 5 = -5 \times 3 = -15$$

$$\underline{\underline{-26}}$$

$$\sum = -12$$

$$\theta_1 = \theta - 0.1 \times (-25)$$

$$0 - 0.1 \times -25 = 2.6$$

$$\theta_1 = 2.6$$

$$\theta_2 = \theta - \alpha \sum_{i=1}^3 (h_{\theta}(x^{(i)}) - y^{(i)}) \cdot x_2^{(i)}$$

$$\sum_{i=1}^3 (h_{\theta}(x^{(i)}) = (\theta_0 + 1\theta_1 + 2\theta_2) - 3 = -3 \times 2 = -6$$

$$h_{\theta}(x^{(2)}) = (\theta_0 + 2\theta_1 + 1\theta_2) - 4 = -4 \times 1 = -4$$

$$h_{\theta}(x^{(3)}) = (\theta_0 + 3\theta_1 + 3\theta_2) - 5 = -5 \times 3 = -15$$

$$\underline{\underline{-25}}$$

$$\theta_2 = 0 - 0.1 \times (-25) = 2.5$$

$$\frac{1}{2} [\nabla_{\theta} (\theta^T X^T X \theta) - \nabla_{\theta} (y^T X \theta) + 0]$$

$$\frac{1}{2} [\nabla_{\theta} (\theta^T X^T X \theta) - \nabla_{\theta} (2 \boxed{X^T y})^T \theta + 0]$$

$$\frac{1}{2} [\nabla_{\theta} (\theta^T \boxed{X^T X} \theta) - 2 X^T y]$$

gradient with respect to θ . identity

$$\nabla_{\theta} X^T A X = 2 A X$$

identity matrix calculus.

$$= \frac{1}{2} [2 X^T X \theta - 2 X^T y]$$

$$\nabla_{\theta} J(\theta) = \frac{1}{2} [2 X^T X \theta - 2 X^T y]$$

set gradient = 0 and find θ .

$$0 = X^T X \theta - X^T y$$

$\downarrow$
 $\Rightarrow 0$. set it to 0

$$X^T X \theta - X^T y = 0$$

$$X^T y = X^T X \theta$$

$$\theta = (X^T X)^{-1} X^T y$$

instead of, division its inverse

$$\boxed{\theta = (X^T X)^{-1} X^T y}$$

challenge while using this

- $\rightarrow$ inverse may not exist
- $\rightarrow$ computational complexity by computing inverse & multiplication is expensive.

directly dependent matrix with inverse problem.

when inverse does not exist, they use pseudoinverse.

small sample size is good.

Example:

x_0	x_1	x_2	y
1	1	2	5
1	3	4	11

$$\theta = (X^T X)^{-1} X^T y$$

$$X = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}_{2 \times 2} \quad y = \begin{bmatrix} 5 \\ 11 \end{bmatrix} \quad \theta = \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}_{2 \times 2}$$

$$X^T X = \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}_{2 \times 2} \quad \begin{aligned} 1(1) + 3(3) &= 10 \\ 1(2) + 3(4) &= 2 + 12 = 14 \\ 2(1) + 4(3) &= 2 + 12 = 14 \\ 2(2) + 4(4) &= 4 + 16 = 20 \end{aligned}$$

$$X^T X = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

$$(X^T X)^{-1}$$

$$(AB)^T = B^T A^T$$

Formula for inverse

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix}$$

$$|A| = (10 \times 20) - (14 \times 14)$$

$$|A| = 200 - 196$$

$$\frac{1}{|A|} = \frac{1}{4}$$

$$x_1^{(1)} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$x_1^{(2)} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

$$y = \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

$$y^T = \begin{bmatrix} 5 & 11 \end{bmatrix}$$

$$\begin{array}{r} 14 \\ 14 \\ \hline 56 \\ 140 \\ \hline 196 \end{array}$$

$$(X^T X)^{-1} = \frac{1}{4} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 10 & 14 \\ 14 & 20 \end{bmatrix} \quad A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{4} \begin{bmatrix} 20 & -14 \\ -14 & 10 \end{bmatrix}$$

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$= \begin{bmatrix} 5 & -3.5 \\ -3.5 & 2.5 \end{bmatrix}$$

Independent columns on a matrix $\rightarrow$ linearly independent

$$\text{Ex: } \begin{bmatrix} 3 & 1 \\ 5 & 2 \end{bmatrix}$$

$$(X^T X)^{-1} X^T$$

dependent columns in a matrix $\rightarrow$ linearly dependent

$$(X^T X)^{-1} X^T$$

$$\begin{bmatrix} 5 & -3.5 \\ -3.5 & 2.5 \end{bmatrix} \begin{bmatrix} 1 & 3 \\ 2 & 4 \end{bmatrix}$$

$$\text{Ex: } \begin{bmatrix} x_1 & x_2 \\ 3 & 6 \\ 6 & -12 \end{bmatrix}$$

$$5(1) + -3.5(2) = 5 - 7 = -2$$

$$2.5(3) + -3.5(4) = 7.5 - 14 = -6.5$$

$$2.5(3.5(1) + 2.5(2) = 7.5 + 5 = 12.5$$

$$-3.5(3.5(3) + 2.5(4) = -10.5 + 10 = -0.5$$

computing success for linearly dependent features is difficult.

$$\begin{bmatrix} -2 & 1 \\ 1.5 & -0.5 \end{bmatrix} \begin{bmatrix} 5 \\ 11 \end{bmatrix}$$

Linear Regression (two matrix perspective)

$$-2(5) + 1(11) = -10 + 11 = 1$$

$$1.5(5) + (-0.5)(11) = 7.5 - 5.5 = 2$$

- gradient descent approach
- closed form solutions

3rd perspective

- probabilistic view point. (math concept prob/stat).

problem

	x_1	x_2	y
x^1	1	0	0
x^2	2	1	0
x^3	3	1	0
x^4	4	2	1
x^5	5	3	1

$$\theta_1 = 0$$

$$\theta_2 = 0$$

$$\alpha = 0.1$$

$$\text{threshold} = 0.5$$

update θ values of 1st iteration.

$$\theta_1 = \theta_1 + \alpha \sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_j^{(i)}$$

$$\sum_{i=1}^5 \left(y - \frac{1}{1 + e^{-\theta^T x^{(i)}}} \right) x_1$$

$$(0 - 0.5) \times 1 = -0.5$$

$$i=2 \left(y - \frac{1}{1 + e^{-\theta^T x^1}} \right) x_2$$

$$(0 - 0.5) \times 2 = -1$$

$$i=3 (0 - 0.5) \times 3 = -1.5$$

$$i=4 (1 - 0.5) \times 4 = 0.5 \times 4 = 2$$

$$i=5 (1 - 0.5) \times 5 = 0.5 \times 5 = 2.5$$

$$-0.5 - 1 - 1.5 + 2 + 2.5 = -3 + 4.5 = 1.5$$

$$\theta_1 = 0 + 0.1 (1.5)$$

$$\theta_0 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \quad \theta^T = [0 \ 0]$$

$$\theta^T x^1 = [0 \ 0] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0$$

$$\theta^T x^2 = [0 \ 0] \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 0$$

$$\theta^T x^3 = [0 \ 0] \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 0$$

$$\theta^T x^4 = [0 \ 0] \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 0$$

$$\theta^T x^5 = [0 \ 0] \begin{bmatrix} 5 \\ 3 \end{bmatrix} = 0$$

$$\theta_2 = \theta_2 + \alpha \sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_2^{(i)}$$

$$\sum_{i=1}^5 \left(y - \frac{1}{1 + e^{-\theta^T x^{(i)}}} \right) x_2$$

$$(0 - 0.5) \times 0 = 0$$

$$i=2 (0 - 0.5) \times 1 = -0.5$$

$$i=3 (0 - 0.5) \times 1 = -0.5$$

$$i=4 (1 - 0.5) \times 2$$

$$0.5 \times 2 = 1$$

$$i=5 (1 - 0.5) \times 3$$

$$0.5 \times 3 = 1.5$$

$$-0.5 - 0.5 + 1 + 1.5 = 1.5$$

$$\theta_2 = 0 + 0.1 (1.5)$$

$$= 0.15$$

$$\theta = \begin{bmatrix} 0.15 \\ 0.15 \end{bmatrix}$$

Second iteration

$$\theta_1^T = [0.15, 0.15]$$

$$\theta_1^T x^1 = [0.15 \ 0.15] \begin{bmatrix} 1 \\ 0 \end{bmatrix} = 0.15(1) + 0.15(0) = \underline{0.15}$$

$$\theta_1^T x^2 = [0.15 \ 0.15] \begin{bmatrix} 2 \\ 1 \end{bmatrix} = 0.15(2) + 0.15(1) = \underline{0.45}$$

$$\theta_1^T x^3 = [0.15 \ 0.15] \begin{bmatrix} 3 \\ 1 \end{bmatrix} = 0.15(3) + 0.15(1) = \underline{0.6}$$

$$\theta_1^T x^4 = [0.15 \ 0.15] \begin{bmatrix} 4 \\ 2 \end{bmatrix} = 0.15(4) + 0.15(2) = \underline{0.9}$$

$$\theta_1^T x^5 = [0.15 \ 0.15] \begin{bmatrix} 5 \\ 3 \end{bmatrix} = 0.15(5) + 0.15(3) = \underline{1.2}$$

$$\theta_1 := \theta_1 + \alpha \sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_i^{(i)}$$

$$\sum_{i=1}^5 (y - \frac{1}{1+e^{-0.15}}) x_1 = (0 - 0.53) \times 1 = -0.53$$

$$i=2 \quad (0 - 0.61) \times 2 = -1.22$$

$$i=3 \quad (0 - 0.64) \times 3 = -1.92$$

$$i=4 \quad (0 - 0.71) \times 4 = -2.84$$

$$i=5 \quad (0 - 0.76) \times 5 = -3.8$$

$$-0.53 - 1.22 - 1.92 + 1.16 + 1.2 = -3.67 + 2.36 = \underline{-1.31}$$

$$\theta_1 := \theta_1 + 0.1(-1.31)$$

$$= 0.15 + 0.1(-1.31)$$

$$= 0.15 + (-0.131)$$

$$= 0.019$$

$$1. \theta_1 = 0.15 + \alpha \left[\frac{\partial L}{\partial \theta_1} \right]$$

$$\text{To find } \frac{\partial L}{\partial \theta_1}$$

$$\frac{\partial L}{\partial \theta_1} = \sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_1^{(i)}$$

To find $h_{\theta}(x^{(i)})$ for i from 1 to 5

$$h_{\theta}(x^{(1)}) = \frac{h_{\theta}(x^{(5)})}{h_{\theta}(x^{(5)})} = \frac{1}{1+e^{-0.15}} = 0.53$$

$$\theta_2 := \theta_2 + \alpha \sum_{i=1}^5 (y^{(i)} - h_{\theta}(x^{(i)})) x_2^{(i)}$$

$$\sum_{i=1}^5 (y - \frac{1}{1+e^{-0.15}}) x_1 = (0 - 0.53) \times 0 = 0$$

$$i=2 \quad (0 - 0.61) \times 1 = -0.61$$

$$i=3 \quad (0 - 0.64) \times 1 = -0.64$$

$$i=4 \quad (1 - 0.71) \times 2 = 0.29 \times 2 = 0.58$$

$$i=5 \quad (1 - 0.76) \times 3 = 0.24 \times 3 = 0.72$$

$$-0.61 - 0.64 + 0.58 + 0.72 = 0.05$$

$$\theta_2 := \theta_2 + 0.1(0.05)$$

$$= 0.15 + 0.1(0.05)$$

$$= 0.15 + 0.005$$

$$= \underline{0.155}$$

$$\theta_2 = \begin{bmatrix} 0.019 \\ 0.155 \end{bmatrix}$$

Problem: predict the risk for autism.

problem 2: Regression tree.
House prediction (price)

IQ Score	Social score	Verbal score	Risk
100	79	86	Low
30	40	70	High
80	85	65	low
90	82	40	High

Area	Bedroom	price
1700	4	60
1500	2	30
1200	3	40
1100	1	20

Test sample.

(1)
 $x_{\text{test}} = \text{IQ} = 120$

Social = 60

Verbal = 40

Risk for autism — ?

Test data

predict the house price for

Area = 1200, Bedroom = 2.

parameters

split dimension

split value

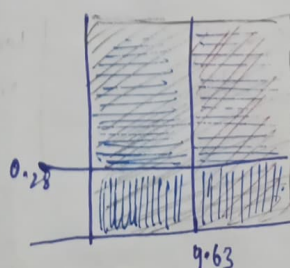
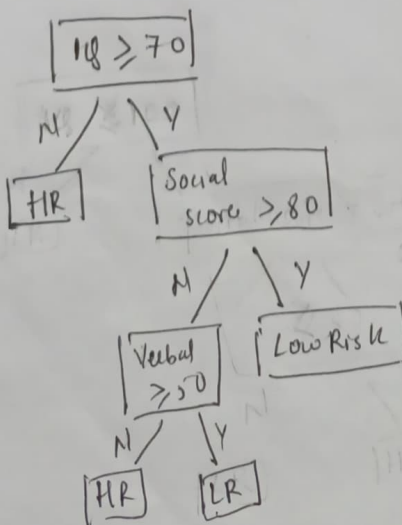
child nodes

• recursive fun when trying every feature to construct an appropriate node.

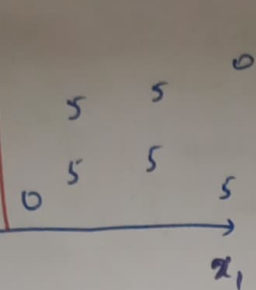
trial:

Labels — Low

— High



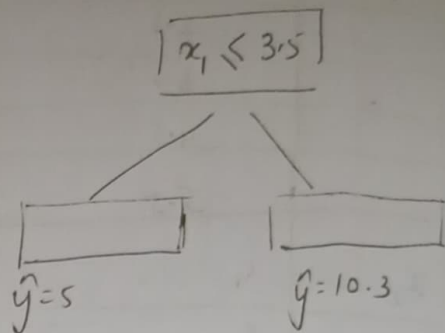
diff sections divided based on split values



btw 0 to 6
there can be
infinite values.
take mean/
average.

$$\text{mean} = \frac{9+10+12}{3} = 10.3$$

There, there is
going to be
some error
while predicting
b/c there is
a diff b/w
ground truth
and predicted.



The model is going to predict
all samples that gets LHS as
5 & RHS as 10.3.

$$S_1 \quad y^{(1)} = 5$$

$$S_2 \quad y^{(2)} = 4$$

$$S_3 \quad y^{(3)} = 6$$

$$\text{MSE}_L = \frac{(5-5)^2 + (4-5)^2 + (6-5)^2}{3} = \boxed{0.67}$$

$$\text{MSE}_R = \frac{(9-10.3)^2 + (10-10.3)^2 + (12-10.3)^2}{3} = \frac{1.69 + 0.09 + 2.89}{3} = \frac{4.67}{3}$$

$$= \boxed{1.56}$$

$$\text{Total mean squared Error MSE} = \text{MSE}_L + \text{MSE}_R = 0.67 + 1.56 = \boxed{2.23}$$

26 Build a regression tree.

	x_2	x_3	y
	3	1	5
	2	2	4
	1	2	6
	3	3	9
	2	1	10
	1	2	12

Take feature x_1 ,
Sort feature values.

[1, 2, 3, 4, 5, 6]

candidate (potential) split points.

[1.5, 2.5, 3.5, 4.5, 5.5]

mean btw two points after being
sorted.

assume candidates 1.5, 2.5.

for x_1 ; candidate split value 3.5

$$x_1 \leq 3.5$$

Sample 1
Sample 2
Sample 3

2, 1, 3
5, 4, 6

Sample 4
Sample 5
Sample 6

4, 5, 6
9, 10, 12

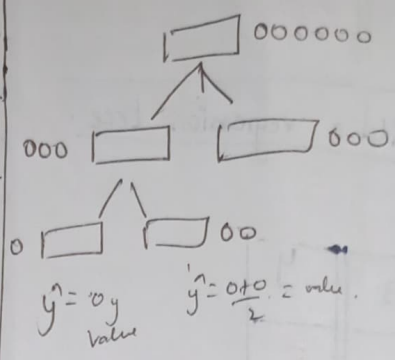
$$\text{mean} = \frac{5+4+6}{3}$$

$$= \frac{15}{3} = \boxed{5}$$

MSE = (5-5)²
Average of all ground
truth values
prediction 5

Feature	Candidate split value	Loss
x_1	1.5	l_1
	2.5	l_2
	3.5	l_3 2.2
	4.5	l_4
	5.5	l_5
x_2	c_1	l_1
	c_2	l_2
	c_3	l_3
	c_4	l_4
	c_5	l_5
x_3		

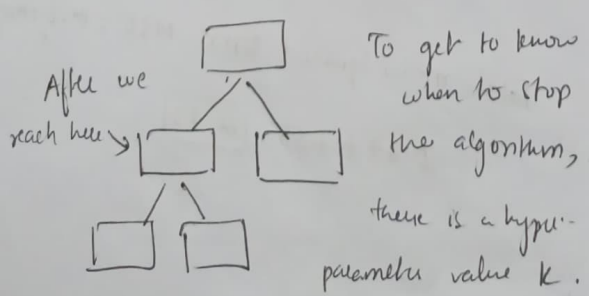
Again take average,
when there is only k values in a
then find the average y of k so
and predict that has $\hat{y}$.



Then choose the pair of candidate split value that has the minimum loss value and place it as root node in the tree.

• if x_2 is sorted
[1, 1, 2, 2, 3, 3]

potential mean $\rightarrow$ 1.5, 2.5



that says how many samples should be in the leaf. Ex: $k=2$

leaf should have 2 samples. If more again divide into

Decision Tree classification

1. Information Gain / Gini Entropy
2. Gini Index
3. Binary cross entropy loss

pruning

$$E(T) = \sum_{i=1}^n E(T(x^i, y^i)) + \lambda |T|$$

$|T| = \text{no. of leaves}$

if no. of leaves $\uparrow$ cost $\uparrow$
depth is $\uparrow$. - overfitting

if no. of leaves $\downarrow$ cost $\downarrow$ depth is $\downarrow$
optimization $\checkmark$

if λ is very low $\rightarrow$ underfitting

x_1	x_2	y
1	1	2
1	2	3
2	2	4
3	2	6
3	3	7

1. feature 1 $\rightarrow x_1$

$[1, 1, 2, 3, 3]$ sorted

Candidate split points

1.5, 2.5

middle blue 1.5

for $x_1 \leq 1.5$

sample 1
2
sample 3
4
5

$$y^{(1)} = 2$$

$$y^{(2)} = 3$$

$$\frac{2+3}{2} = 2.5$$

prediction

$$y^{(3)} = 4$$

$$y^{(4)} = 6$$

$$y^{(5)} = 7$$

$$\frac{4+6+7}{3} = 5.6$$

prediction

$$MSE = \frac{(2-2.5)^2 + (3-2.5)^2}{2}$$

$$= \frac{0.25 + 0.25}{2} = 0.25$$

$$MSE = \frac{(4-5.6)^2 + (6-5.6)^2 + (7-5.6)^2}{3}$$

$$= \frac{2.56 + 0.16 + 1.96}{3} = 1.56$$

Feature	Candidate split value	Loss
x_1	1.5	1.81
	2.5	0.9
x_2	1.5	2.5
	2.5	2.185

for $x_1 \leq 2.5$

sample 1
sample 2
sample 3

$$y^{(1)} = 2$$

$$y^{(2)} = 3$$

$$y^{(3)} = 4$$

$$\frac{2+3+4}{3} = 3$$

$$\hat{y} = 3$$

$$MSE_1 = \frac{(2-3)^2 + (3-3)^2 + (4-3)^2}{3}$$

$$= \frac{2}{3} = 0.66$$

sample 4
sample 5

$$y^{(4)} = 6$$

$$y^{(5)} = 7$$

$$\frac{6+7}{2} = 6.5$$

$$\hat{y} = 6.5$$

$$MSE_2 = \frac{(6-6.5)^2 + (7-6.5)^2}{2}$$

$$= 0.25$$

for x_2

[1, 2, 2, 2, 3]

candidate split point

1.5, 2.5

for $|x_2 \leq 1.5|$

sample 1

sample 2

$y(2) = 3$

$y(3) = 3$

$y(4) = 6$

$y(5) = 7$

$y^{(1)} = 2$

$\hat{y} = 2$

$\frac{3+3+6+7}{4} = 5$

$\hat{y} = 5$

$MSE_1 = \frac{(2-2)^2}{1} = 0 = 0$

$MSE_2 = \frac{(3-5)^2 + (3-5)^2 + (6-5)^2 + (7-5)^2}{4} = \frac{4+1+1+4}{4} = \frac{10}{4} = 2.5$

$2.5 + 0 = 2.5$

for $|x_2 \leq 2.5|$

sample 1

$y(1) = 2$

$y(2) = 3$

$y(3) = 4$

$y(4) = 6$

sample 5

$y(1) = 7$

$\frac{2+3+4+6}{4} = \frac{15}{4} = 3.75$

$\frac{7}{1} = 7$

$(2-3.75)^2 + (3-3.75)^2 + (4-3.75)^2 + (6-3.75)^2$

$= \frac{3.06 + 0.56 + 0.06 + 5.06}{4} = 2.185$

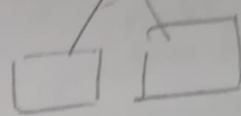
$|x_1 \leq 2.5|$

sample 2 $|x_1 \leq 1.5|$

sample 4

$\hat{y}$ = average of y values

report if leaf node



sample 1

sample 2

3

x_1	x_2	y
1	1	2
1	2	3
2	2	4

for

for x_1

[1, 1, 2]

candidate split point = 1.5

$|x_1 \leq 1.5|$

sample 1

3

$y^{(1)} = 2$

$y^{(2)} = 3$

$y^{(3)} = 4$

$\frac{4}{1} = 4$

$\frac{2+3}{2} = \frac{5}{2} = 2.5$

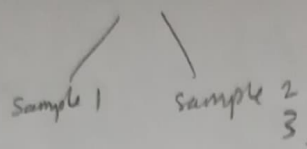
$MSE_1 = \frac{(2-2.5)^2 + (3-2.5)^2}{2}$

$= 0.25 + 0.25 = 0.5$

$MSE_2 = 0$

$MSE = 0.5 + 0 = 0.5$

for $x_2 \leq 1.5$



$$y^{(1)} = 2$$

$$y = 2$$

$$(2-2)^2 = 0$$

$$MSE_1 = 0$$

$$y^{(2)} = 3$$

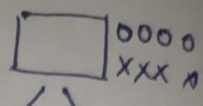
$$y^{(3)} = 4$$

$$\frac{3+4}{2} = 3.5$$

$$MSE_2 = (3-3.5)^2 + (4-3.5)^2$$

entropy ↑
random
chaotic

heterogeneous nodes.



pure
entropy ↓

homogeneous nodes

so entropy is used to construct tree features.

Decision Tree Classification

0.5

Entropy, H (Shannon's info theory)

$$H_{\text{root}} = \sum_{i=1}^2 -P_i \log_2 P_i$$

$$= -P_{\text{yes}} \log_2 P_{\text{yes}}$$

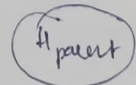
$$+ (-P_{\text{no}} \log_2 P_{\text{no}})$$

$$= -0.5 \log_2 (0.5) + (-0.5) \log_2 (0.5)$$

$$= -0.5 \log_2 (0.5) +$$

$$-0.5 \log_2 (0.5)$$

$$H_{\text{root}} = 1$$



entropy ↑ ↑ very
entropy 0 ↓ very

step 1: Compute root entropy

target variable.

$$P(\text{Yes}) = \frac{4}{8} = \frac{2^1}{4^2} = 0.5$$

$$P(\text{No}) = \frac{4}{8} = 0.5$$

Entropy, H (measure of impurity / chaos in data)

if entropy ↑ more impure / randomness / chaos.

if entropy ↓ data is pure.

Step 2. Compute Information gain

IG, for feature = weather

(choose which feature gives you most information gain).

x_1 weather has 3 value.

all unique features - vol

$X_1 \rightarrow$ Weather

Weather has 3 values
sunny, overcast and rain.

$$-P_{yes} \log_2 P_{yes} = -0.6 + 1.2311 = 1.5157$$

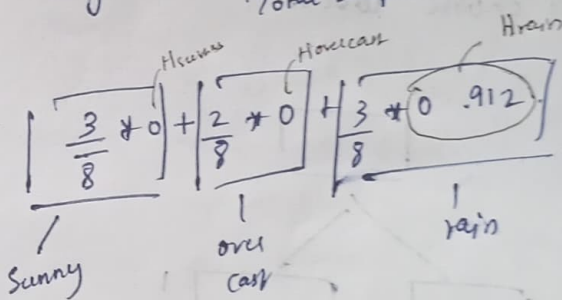
$$-P_{no} \log_2 P_{no} = -0.3 + 1.2311$$

$$= -0.9094 + (-0.36933)$$

$$H_{rain} = 1.912$$

Step 3
Compute weighted entropy

$$H_{weighted} = \frac{\text{no. of sample}}{\text{Total sample}} H_{child}$$



Step 4
Information Gain weather

$$IG_{weather} =$$

$$H_{parent} - H_{child(weighed)}$$

$$= H_{root} - H_{child}$$

$$= 1 - 0.344$$

$$IG_{weather} = 0.656$$

Higher $IG \uparrow$ good. (info is gained)

Step 5
Pick feature that gives the maximum information $IG / \text{Arg max } IG$.

Sunny
3 samples, S_1, S_2, S_8 (No, No, No).

$$P(\text{No}) = \frac{3}{3} = 1, P(\text{Yes}) = \frac{0}{3} = 0$$

$$H_{sunny} = \sum_{i=1}^1 -P_i \log_2 P_i \text{ entropy} = 0, \text{ impurity} = 0$$

$-1 \log_2(1)$ pure homogenous

$$H_{sunny} = 0$$

overcast
2 samples (S_3, S_7) (Yes, Yes)

$$P(\text{Yes}) = \frac{2}{2} = 1$$

$$H_{overcast} = \sum_{i=1}^1 -P_i \log_2 P_i$$

$$-1 \log_2(1)$$

$$= 0$$

overcast

entropy = 0, impurity = 0, pure, homogenous

rain
3 samples (S_4, S_5, S_6) (Yes, No, Yes)

$$P(\text{Yes}) = \frac{2}{3} = 0.6$$

$$i = 2$$

$$P(\text{No}) = \frac{1}{3} = 0.3$$

$$H_{rain} = \sum_{i=1}^2 -P_{Yes} \log_2 P_{Yes} +$$

$$-P_{No} \log_2 P_{No}$$

x_1	x_2	y
1	2	4
2	3	5
3	4	6
4	5	7

$$\lambda = 0.1$$

$$[x_1 \geq 2.5]$$

$$\begin{array}{c} \swarrow \quad \searrow \\ \boxed{4.05} \quad \boxed{5.85} \end{array}$$

Sample 1 2 Sample 3 4

Iteration 1

$$r_1 = y_1$$

(a) Fit a model f_1

$$[x_1 \geq 2.5]$$

Decision stump.

$$\begin{array}{c} \swarrow \quad \searrow \\ \boxed{4.05} \quad \boxed{6.5} \end{array}$$

generate the residuals.

$$r^{(1)}_1 = y^{(1)} - \lambda f_1(x^{(1)})$$

$$= 4 - 0.1(4.5) = 3.55$$

$$r^{(2)}_1 = y^{(2)} - \lambda f_1(x^{(2)})$$

$$= 5 - (0.1 \times 4.5) = 4.55$$

$$r^{(3)}_1 = y^{(3)} - \lambda f_1(x^{(3)})$$

$$6 - (0.1 \times 6.5) = 5.35$$

$$r^{(4)}_1 = y^{(4)} - \lambda f_1(x^{(4)})$$

$$7 - (0.1 \times 6.5) = 6.35$$

x_1	x_2	r
1	2	3.55
2	3	4.55
3	4	5.35
4	5	6.35

$$\begin{aligned} r^{(1)}_2 &= r^{(1)}_1 - \lambda f_2(x^{(1)}) \\ &= 3.55 - (0.1 \times 4.05) \\ &= 3.145 \end{aligned}$$

$$\begin{aligned} r^{(2)}_2 &= r^{(2)}_1 - \lambda f_2(x^{(2)}) \\ &= 4.55 - 0.1 \times 4.05 \\ &= 4.145 \end{aligned}$$

$$\begin{aligned} r^{(3)}_2 &= r^{(3)}_1 - \lambda f_2(x^{(3)}) \\ &= 5.35 - (0.1 \times 5.85) \\ &= 4.765 \end{aligned}$$

$$\begin{aligned} r^{(4)}_2 &= r^{(4)}_1 - \lambda f_2(x^{(4)}) \\ &= 6.35 - 0.585 \\ &= 5.765 \end{aligned}$$

$$f = \lambda f_1 + \lambda f_2 + \lambda f_3 + \dots + \lambda f_n$$

- weak learners
- learn from mistakes of previous learners (residuals).
- Aggregate all models / learners with a weighted factor.
- same λ ,

Adaboost classification

x_1	x_2	y	W
1	2	+1	0.25
2	1	-1	0.25
4	4	+1	0.25
5	3	-1	0.25

(1) Sample weight

(2) model weight

After finding the split values for all features and identifying the feature, & split value that gives least error, construct a decision stump / stub tree.

$$f_t = \underset{t}{\operatorname{argmin}} \sum_i w^{(i)} I(y^{(i)} \neq f(x^{(i)}; \theta))$$

possible split values of $x_1 =$

1.5, 2, 3, 4.5

So

$$|x_1 < 1.5|$$

(Assume this the decision stump)

$$\begin{array}{c} N \quad Y \\ \swarrow \quad \searrow \end{array}$$

$$\begin{array}{c} \boxed{-1} \quad \boxed{+1} \end{array}$$

sample $\frac{3}{4}$ Sample 1

$$e = 0$$

[Error for this sample is 0 bcz I returns 0 when it is true] the $y = +1$, & in $\hat{y}$ also +1]

Ans

$$e^{(0)} = 0$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$$e^{(3)} = 0.25$$

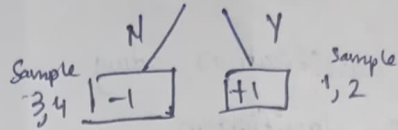
$$\begin{bmatrix} y = +1 \\ \hat{y} = -1 \end{bmatrix}$$

$$e^{(0)} = 0$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$\therefore$ for this split the total error is 1
 $e^1 + e^2 + e^3 + e^4 = 0.25$

$$f_1 \quad |x_1 < 3|$$



$$e^1 = 0$$

$$\begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^2 = 0.25$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^3 = 0.25$$

$$\begin{bmatrix} y = +1 \\ \hat{y} = -1 \end{bmatrix}$$

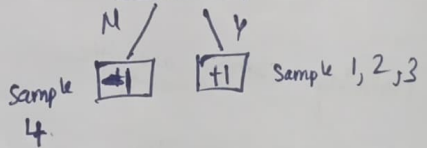
$$e^4 = 0$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$$e^1 + e^2 + e^3 + e^4 = 0.25 + 0.25 = 0.5$$

$\therefore$ for this split the total error is 0.5

$$f_2 \quad |x_1 \leq 4|$$



$$e^1 = 0$$

$$\begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^2 = 0.25$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^3 = 0$$

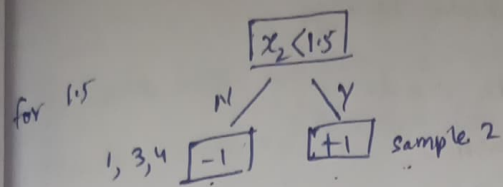
$$\begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^4 = 0$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$\therefore$ for this split the total error is 0.5

For feature x_2 , the split values are 1.5, 2.5, 3.5.



$$e^1 = 0.25 \quad \begin{bmatrix} y = +1 \\ \hat{y} = -1 \end{bmatrix}$$

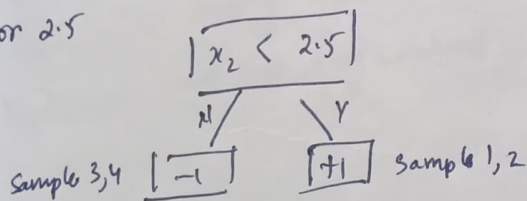
$$e^2 = 0.25 \quad \begin{bmatrix} y = -1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^3 = 0.25 \quad \begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^4 = 0 \quad \begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$$= 0.75$$

for 2.5



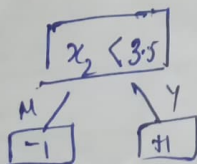
$$e^1 = 0 \quad \begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^2 = 0.25 \quad \begin{bmatrix} y = -1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^3 = 0.25 \quad \begin{bmatrix} y = +1 \\ \hat{y} = -1 \end{bmatrix}$$

$$e^4 = 0 \quad \begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$$= 0.5$$



$$\text{error} = 0.75$$

(b) $\rightarrow$ error that we get by model for f_0 (model 1)

model weight =

$$\lambda_1 = \frac{1}{2} \log_2 \left(\frac{1-e}{e} \right)$$

$$= \frac{1}{2} \ln \left(\frac{1-0.25}{0.25} \right)$$

$$\lambda_1 / \lambda_1 = 0.5493$$

(c) update sample weights.

$$w_1^{(i)} \leftarrow w_1^{(i)} e^{-\lambda_1} \text{ if correct}$$

$$w_1^{(i)} \leftarrow w_1^{(i)} e^{\lambda_1} \text{ if incorrect}$$

	old w	new w	Normalized w
S1	0.25	0.144	0.166
S2	0.25	0.144	0.166
S3	0.25	0.43	0.500
S4	0.25	0.144	0.166
Total		0.862	

$$0.25 \times e^{-(0.5493)} = 0.144$$

$$0.25 \times e^{-0.5493} = 0.144$$

$$0.25 \times e^{+0.5493} = 0.43$$

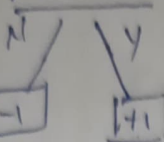
$$0.25 \times e^{-0.5493} = 0.144$$

$$\text{normalized} = \frac{\text{S1 new weight}}{\sum \text{sum of all new weights}}$$

— iteration 2

Iteration 2

$$|x < 1.5|$$



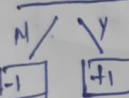
	x_1	x_2	y	w
1	2	1	-1	0.33
2	4	4	+1	0.33
3	5	3	-1	0.33

(a) all possible split values for feature

x_1

3, 4.5

$$|x_1 < 3|$$



$$e^{(1)} = 0.33$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = 1 \end{bmatrix}$$

$$e^{(2)} = 0.33$$

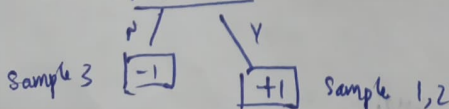
$$\begin{bmatrix} y = +1 \\ \hat{y} = -1 \end{bmatrix}$$

$$e^{(3)} = 0$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

$$\therefore e = 0.66$$

$$|x_1 < 4.5|$$



$$e^1 = 0.33$$

$$\begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^2 = 0$$

$$\begin{bmatrix} y = +1 \\ \hat{y} = +1 \end{bmatrix}$$

$$e^3 = 0$$

$$\therefore e = 0.33$$

$$\begin{bmatrix} y = -1 \\ \hat{y} = -1 \end{bmatrix}$$

ABDM

Ayushman Bhanja

→ health records

→ laboratories

→ technicians

→ API's to access these datasets

Gradient Boosting for classification

Trial these as prob values.

chest pain	Good Blood circulation	Blocked arteries	Heart disease	residual
no	no	no	no → 0	$0 - 0.66 = -0.66$
yes	yes	yes	yes → 1	$1 - 0.66 = 0.34$
yes	yes	no	yes → 1	$1 - 0.66 = 0.34$
yes	no	no	yes → 1	$1 - 0.66 = 0.34$
yes	no	yes	yes → 1	$1 - 0.66 = 0.34$
no	yes	no	no → 0	$0 - 0.66 = -0.66$

no → 0 } in target
yes → 1

odds = 4/2.

$-0.6 \sim -0.7$
 $+0.34 \sim 0.3$

- (1) log-odds
- (2) convert the prob

step 1: initial prediction

$$\log(\text{odds}) = \log\left(\frac{4}{2}\right) \approx 0.7$$

step 2: convert this to probability

$$p = \frac{e^{\log(\text{odds})}}{1 + e^{\log(\text{odds})}}$$

$$= \frac{e^{\log(4/2)}}{1 + e^{\log(4/2)}} \approx 0.66$$

i.e. all samples sitting in that node/leaf.

for first leaf

all values in that leaf

$$\sum_i \text{Residual}_i$$

$$\sum \text{Previous prob}_i \times (1 - \text{Prev Prob})$$

$$\text{1st leaf} = \frac{0.3}{(0.66) \times (1 - 0.66)} \approx 1.43$$

for second leaf

2nd leaf =

$$\sum (0.3) + (-0.7)$$

$$\sum (0.66) \times (1 - 0.66) + (0.66) \times (1 - 0.66)$$

$$\approx -0.95$$

for third leaf

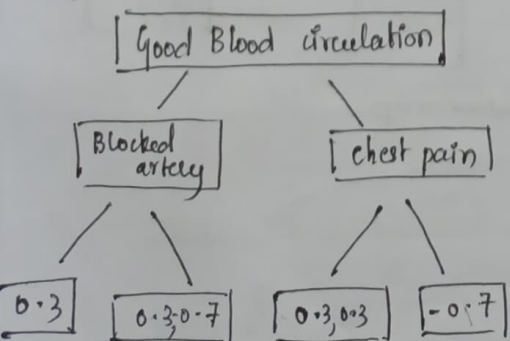
3rd leaf

$$= 1.42$$

for 4th leaf =

$$-3.33$$

Imagine we have constructed this tree.



$$a = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

$$z = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$$

$$\begin{matrix} 1 \times 4 & 4 \\ 2 \times 5 & = 10 \\ 3 \times 6 & 18 \end{matrix}$$

Example 2

$$z = \begin{bmatrix} 1 \\ 2 \end{bmatrix}_{2 \times 1} \quad Z = \begin{bmatrix} 3 \\ 4 \end{bmatrix}_{2 \times 1} \quad \mathbb{R}^2$$

$$\phi x = \begin{bmatrix} 1 \times 1 \\ 1 \times 2 \\ 1 \times 3 \\ 2 \times 1 \\ 2 \times 2 \\ 2 \times 3 \\ 3 \times 1 \\ 3 \times 2 \\ 3 \times 3 \end{bmatrix} \rightarrow \begin{bmatrix} 1 \\ 2 \\ 3 \\ 2 \\ 4 \\ 6 \\ 3 \\ 6 \\ 9 \end{bmatrix}_{9 \times 1}$$

prove
to work
or working

$$\phi(x) = \begin{bmatrix} x_1^2 \\ x_2^2 \\ \sqrt{2} x_1 x_2 \\ \sqrt{2} x_1 \\ \sqrt{2} x_2 \\ 1 \end{bmatrix}_{6 \times 1} \quad \mathbb{R}^6$$

$$\phi: \mathbb{R}^2 \rightarrow \mathbb{R}^6$$

$$K(x, z) = (x \cdot z + 1)^2 = 144$$

$$\phi z = \begin{bmatrix} 4 \times 4 \\ 4 \times 5 \\ 4 \times 6 \\ 5 \times 4 \\ 5 \times 5 \\ 5 \times 6 \\ 6 \times 4 \\ 6 \times 5 \\ 6 \times 6 \end{bmatrix} \rightarrow \begin{bmatrix} 16 \\ 20 \\ 24 \\ 20 \\ 25 \\ 30 \\ 24 \\ 30 \\ 36 \end{bmatrix}_{9 \times 1}$$

dot product of ϕx & ϕz will give a single value

$$\phi(x) \cdot \phi(z) = 16 + 40 + 72 + \dots = 1024$$

$$\phi(x)^T \phi(z) =$$

$$K(x, z)$$

Kernel function

$$K(x, z) = (\langle x, z \rangle)^2$$

$$K(x, z) = (4 + 10 + 18)^2$$

$$= 1024$$

value from higher dimensional space can be gotten just by doing something on the original n dimensional space.

$$p \gg d$$

p is sufficiently greater than d .

$$\phi x = \begin{bmatrix} 1^2 \\ 2^2 \\ \sqrt{2} \times 1 \times 2 \\ \sqrt{2} \times 1 \\ \sqrt{2} \times 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 4 \\ 2.82 \\ 1.41 \\ 2.82 \\ 1 \end{bmatrix}$$

$$\phi z = \begin{bmatrix} 3^2 \\ 4^2 \\ \sqrt{2} \times 3 \times 4 \\ \sqrt{2} \times 3 \\ \sqrt{2} \times 4 \\ 1 \end{bmatrix} = \begin{bmatrix} 9 \\ 16 \\ 16.97 \\ 4.24 \\ 5.65 \\ 1 \end{bmatrix}$$

dot product $\phi x \cdot \phi z$

$$(1 \times 9) + (4 \times 16) + (2.82 \times 16.97) + (1.41 \times 4.24) + (2.82 \times 5.65) + (1 \times 1)$$

$$= 9 + 64 + \dots$$

$$= 143.76$$

$$((3+8)+1)^2 = (11+1)^2 = 144$$

$$K(x, z) = (x \cdot z + 1)^2 = 144$$

$$\phi(x)^T \phi(z) = 144$$

Problem

Test data

Actual label	Predicted score	Predicted label	
1	0.85	1	TP
0	0.6	1	
1	0.7	1	TP
1	0.4	0	
0	0.55	1	
1	0.5	1	TS
0	0.65	1	
0	0.35	0	TN
1	0.6	1	TP
0	0.2	0	TN

$N=10$

- Compute:
- (1) Confusion matrix
 - (2) Accuracy
 - (3) Precision
 - (4) Recall / sensitivity
 - (5) F1-Score
 - (6) ROC-curve
- (0.7, 0.5, 0.4, 0.2)

Assume

Threshold = 0.5

Confusion matrix

	original P	original N
predictive +ve	TP → 4	FP → 3
predictive -ve	FN → 1	TN → 2

Accuracy → $\frac{(TP+TN)}{\text{Total}}$

$$\frac{4+2}{10} = \frac{6}{10} = 0.6$$

precision → $\frac{TP}{(TP+FP)}$

$$= \frac{4}{(4+3)} = 0.57$$

Recall / Sensitivity = $\frac{TP}{(TP+FN)}$

$$= \frac{4}{(4+1)} = 0.8$$

F1-Score

$$2 \times \frac{\text{precision} \times \text{recall}}{\text{precision} + \text{recall}} = 2 \times \frac{0.456}{1.37}$$

$$= \frac{1}{\frac{1}{\text{precision}} + \frac{1}{\text{recall}}}$$

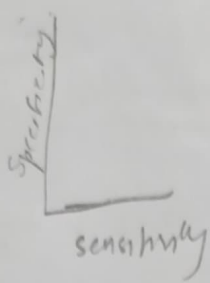
F1 →

$$2 \times 0.32 = 0.65$$

specificity

$$\frac{TN}{(TN+FP)} = \frac{2}{(2+3)} = \frac{2}{5} = 0.4$$

Roc curve



Threshold 0.2, 0.4, 0.5, 0.7

0.2 TP → 5 FP → 5

FN → 0 TN → 1

0.2

Actual label	Predictive score	Predicted label	
1	0.85	1	TP
0	0.6	1	FP
1	0.7	1	TP
1	0.4	1	TP
0	0.35	1	FP
1	0.65	1	TP
0	0.5	1	FP
0	0.35	1	FP
1	0.6	1	TP
0	0.2	0	TN

$$\text{Specificity} = \frac{TN}{(TN+FP)}$$

$$\text{Sensitivity} = \frac{TP}{(TP+FN)}$$

$$\text{Specity} = \frac{1}{(5+5)} = \frac{1}{10} = 0.1$$

$$\text{Sensitivity} = \frac{5}{(5+0)} = \frac{5}{5} = 1$$

Threshold 0.4

TP - 5 FP - 3

FN - 0 TN - 2

Actual label	Predictive score	Predicted label	
1	0.85	1	TP
0	0.6	1	FP
1	0.7	1	TP
1	0.4	1	TP
0	0.35	1	FP
1	0.65	1	TP
0	0.5	1	FP
0	0.35	0	TN
1	0.6	1	TP
0	0.2	0	TN

$$\text{Specificity} = \frac{TN}{(TN+FP)} = \frac{2}{2+3} = \frac{2}{5}$$

$$\text{Sensitivity} = \frac{TP}{(TP+FN)} = \frac{5}{5+0} = 1$$

threshold 0.7

Actual label	Predictive score	Predicted label	
1	0.85	1	TP
0	0.6	0	TN
1	0.7	1	TP
1	0.4	0	FN
0	0.35	0	TN
1	0.65	0	FN
0	0.5	0	TN
0	0.35	0	TN
1	0.6	0	FN
0	0.2	0	TN

$$\text{Specificity} = \frac{TN}{(TN+FP)} = \frac{5}{5+0} = 1$$

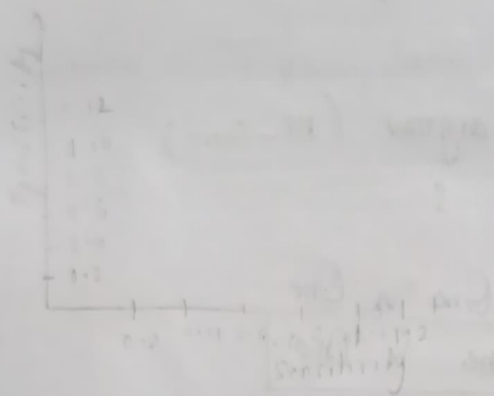
TP - 2 FP - 0

FN - 2 TN - 5

$$\text{Sensitivity} = \frac{TP}{(TP+FN)} = \frac{2}{2+2} = \frac{2}{4} = 0.5$$

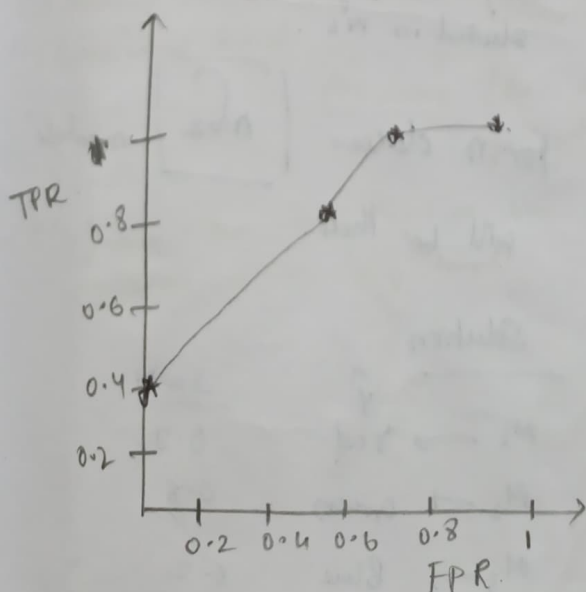
Threshold	Specificity	Sensitivity
0.2	0.1	1
0.4	0.4	1
0.5	0.4	0.8
0.7	1	0.4

ROC curve



$$FPR = 1 - \text{Specificity}$$

Threshold	FPR	TPR (Sensitivity)
0.2	0.9	1
0.4	0.6	1
0.5	0.6	0.8
0.7	0	0.4



$$O^T A u$$



To find $u \rightarrow$ eigen vectors.
sol

known as eigen decomposition problem.

A's eigen vectors will give value of u and eigen value λ , correlation $\rightarrow$ if one quantity increases the other also increases.

every eigen value $\rightarrow$ eigen vector.

rank the eigen value.

λ_1 u_1 ———
 λ_2 u_2 ———
 λ_3 u_3 ———
 eigen value eigen vectors Subspaces.

dimension $\rightarrow$ k dimension
 $d \gg k$
 features will be reduced.

x	$x - \bar{x}$	$(x - \bar{x})^2$
1	-2	4
2	-1	1
3	0	0
4	1	1
5	2	4

y	$y - \bar{y}$	$(y - \bar{y})^2$
2	-2	4
4	0	0
5	1	1
4	0	0
5	1	1

$\bar{x} = 3$
 $\sum_{i=1}^n x - \bar{x} = 10$

$\bar{y} = \frac{20}{5} = 4$
 $\sum (y - \bar{y})^2 = 6$

Correlation & Covariance

x	y
1	2
2	4
3	5
4	4
5	5

x: Hours studied

y: Exam score

compute covariance
correlation

$$r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

(2) Covariance $\rightarrow$

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

$$(3) \sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2}$$

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x^{(i)}$$

$$\bar{x} = \frac{1}{n} \sum (1+2+3+4+5) = \frac{15}{5} = 3$$

$$\sigma_x = \sqrt{\frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})^2} = \sigma_x = \sqrt{\frac{1}{5}(10)} = \sqrt{2}$$

$$\text{cov}(x, y) = \frac{1}{n} \sum_{i=1}^n (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})$$

$$\begin{aligned} \frac{1}{5} \sum_{i=1}^n -2 \times -2 &= \frac{1}{5} \times 4 \\ &= -1 \times 0 = \frac{1}{5} \times 0 = 0 \\ &= 0 \times 1 = 0 \\ &= 1 \times 0 = 0 \\ &= 2 \times 1 = \frac{1}{5} \times 2 = \frac{2}{5} \\ &= \frac{1}{5} \times 6 = \frac{6}{5} \end{aligned}$$

$$\sigma_y = \sqrt{\frac{1}{n} \sum_{i=1}^n (y^{(i)} - \bar{y})^2}$$

$$= \sqrt{\frac{1}{5} \sum 6} = \sqrt{1.2}$$

1	2
2	1
3	2
4	3
5	4

$$\text{correlation} = r = \frac{\text{cov}(x, y)}{\sigma_x \sigma_y}$$

$$= \frac{1.2}{\sqrt{2} \times \sqrt{1.2}} = \frac{1.2}{1.54} = \underline{\underline{0.77}}$$

Eigen values and Eigen vectors.

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\det(A - \lambda I) = 0$$

$$\lambda_1 = 3 \quad \lambda_2 = 1$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 0$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 3 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} - 1 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

For 2D, feature 1 + feature 2
= 100%
variance

For 1D, feature 1 only
= 73% of the data is
only explained

PCA problem:

Reduce the dimensionality from

(a) 3D $\rightarrow$ 2D

(b) 3D $\rightarrow$ 1D. mention covariance.

$$X = \begin{bmatrix} x_1 & x_2 & x_3 \\ 2 & 0 & 1 \\ 0 & 2 & 1 \\ 3 & 1 & 2 \\ 1 & 3 & 2 \end{bmatrix} \quad d=3.$$

$\frac{6}{4} \quad \frac{6}{4} \quad \frac{6}{4} = 1.5$

step 1: Mean centering

$$\mu = [1.5 \quad 1.5 \quad 1.5]$$

Subtract every value from a feature with its respective mean.

$$X_c = \begin{bmatrix} x_1 & x_2 & x_3 \\ 0.5 & -1.5 & -0.5 \\ -1.5 & 0.5 & -0.5 \\ 1.5 & -0.5 & 0.5 \\ -0.5 & 1.5 & 0.5 \end{bmatrix} \quad \text{mean centered values.}$$

step 2 Covariance matrix.

$$\begin{bmatrix} \text{cov}(x_1, x_1) & \text{cov}(x_1, x_2) & \text{cov}(x_1, x_3) \\ \text{cov}(x_2, x_1) & \text{cov}(x_2, x_2) & \text{cov}(x_2, x_3) \\ \text{cov}(x_3, x_1) & \text{cov}(x_3, x_2) & \text{cov}(x_3, x_3) \end{bmatrix}$$

$$C = \frac{1}{n} X_c^T X_c = \begin{bmatrix} 1.25 & -0.75 & 0.5 \\ -0.75 & 1.25 & 0.5 \\ 0.5 & 0.5 & 0.25 \end{bmatrix}$$

step 3 compute eigen values and eigen vectors of covariance matrix

$$\lambda_1 \approx 2.0$$

$$\lambda_2 \approx 0.75$$

$$\lambda_3 \approx 0.0$$

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} \quad u_2 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \quad u_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix}$$

step 4: dimensionality reduction.

• 3D $\rightarrow$ 2D

Take top two eigen vectors.

$$W_{2D} = [V_1 \quad V_2]$$

• project the data point.

$$Z = X_c W_{2D} = []$$

each point will be now in 2D.

• 3D $\rightarrow$ 1D

Take top 1 eigen vector

$$W_{1D} = [V_1]$$

Total variance

$$= 2 + 0.75 + 0$$

$$= 2.75$$

$$PC1 = \frac{2}{2.75} \approx 73\%$$

$$PC2 = \frac{0.75}{2.75} \approx 27\%$$

$$PC3 = \frac{0}{2.75} \approx 0\%$$

PC1 explains 73% of the data.

PC2 / feature 2 alone explains 27% of the data.

PC1 & PC2 together give 100% data.
So feature 1 & 2 will give 100% data.

	x_1	x_2
s_1	1	1
s_2	1	2
s_3	2	2
s_4	8	8
s_5	8	9
s_6	9	8

$$k=2$$

c_1, c_2

randomly assign
samples

step 1

$$c_1 \rightarrow s_2, s_4, s_6$$

$$c_2 \rightarrow s_1, s_3, s_5$$

step 2. compute the centroid.

cluster 1

$$(1,2)(8,8)(9,8)$$

$$\frac{1+8+9}{3} = 6$$

$$\frac{2+8+8}{3} = 6$$

centroid c_1
 $(6,6)$

cluster 2

$$(1,1)(2,2)(8,9)$$

$$\frac{1+2+8}{3} = 3.66$$

$$\frac{1+2+9}{3} = 4$$

centroid c_2
 $(3.66, 4)$

Sample 1
 $(1,1)$

centroid 2 $(3.66, 4)$

$$\sqrt{(3.66-1)^2 + (4-1)^2} = 4.008$$

Sample 2
 $(1,2)$

centroid 1
 $(6,6)$

centroid 2
 $(3.66, 4)$

$$\sqrt{(6-1)^2 + (6-2)^2} = 6.40$$

$$\sqrt{(3.66-1)^2 + (4-2)^2} = 3.32$$

Sample 3

$(2,2)$

centroid 1
 $(6,6)$

centroid 2
 $(3.66, 4)$

$$\sqrt{(6-2)^2 + (6-2)^2} = 5.65$$

$$\sqrt{(3.66-2)^2 + (4-2)^2} = 2.59$$

Sample 4

$(8,8)$

centroid 1
 $(6,6)$

centroid 2
 $(3.66, 4)$

$$\sqrt{(6-8)^2 + (6-8)^2} = 2.82$$

$$\sqrt{(3.66-8)^2 + (4-8)^2} = 5.90$$

Sample	Centroid 1	Centroid 2
$(1,1)$		
$(1,2)$		
$(2,2)$		

Sample 5

$(8,9)$

centroid 1
 $(6,6)$

centroid 2
 $(3.66, 4)$

$$\sqrt{(6-8)^2 + (6-9)^2} = 3.60$$

$$\sqrt{(3.66-8)^2 + (4-9)^2} = 6.62$$

Steps: For each sample, given the centroids choose which centroid is closer to it.

centroid 1

Sample 1 $(1,1)$ $(6,6)$

$$\sqrt{(6-1)^2 + (6-1)^2} = 7.07$$

(2) Single Linkage

the distance b/w two clusters is defined as the min distance b/w any single element in the first cluster and any single element in the second cluster.

$$d(\{A, B\}, C) = \min(d(A, C), d(B, C))$$

(3) Average Linkage

$$d(\{A, B\}, C) = \frac{d(A, C) + d(B, C)}{2}$$

(4) Centroid Linkage

use centroids of each cluster and compute the dist between cluster centroids.

Problem

4 samples.

A	1
B	2
C	5
D	8

$$R^d = 1$$

distance represented in matrix.

	A	B	C	D
A	0	1	4	7
B	1	0	3	6
C	4	3	0	3
D	7	6	3	0

→ use complete linkage.

Iteration 1

$\{A\} \{B\} \{C\} \{D\}$

$\{A, B\} \{C\} \{D\}$

update the matrix.

	$\{A, B\}$	$\{C\}$	$\{D\}$
$\{A, B\}$	0	4	7
$\{C\}$	4	0	3
$\{D\}$	7	3	0

- use linkage here.

- use complete linkage

merge the one that keeps the least on

A to C → 4 — max

B to C → 3

A to D → 7 — max

B to D → 6

$\{A, B\} \{C, D\}$

	$\{A, B\}$	$\{C, D\}$
$\{A, B\}$	0	7
$\{C, D\}$	7	0

A → C 4

A → D 7 →

B → C 3

B → D 6